Master Thesis Project 1 (ITIT-5199)

SecGuard

Risk-Adaptive Isolation for Profit-Aware 5G Network Slicing

Presented by

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The Core Problem in 5G Network Slicing

- **One Physical Network, Many Users:** 5G uses "Network Slicing" to carve up a single physical network (cell towers, fiber cables, servers) into separate virtual networks for different companies.
- **Completely Different Needs on the Same System:**
  - **Cheap & Basic Users:** Simple devices (smart meters, sensors) that only need an inexpensive, low-bandwidth connection.
  - **Life-Critical Users:** High-priority services (remote robotic surgery, connected ambulances, self-driving cars) that require 100% reliability and strict security.
- **The Danger of Sharing:** Because both types of users run on the exact same physical servers, a cyber-attack on a cheap, unprotected device can spread across the server and compromise the critical service next to it.

The Dilemma: Cutting Costs vs. Staying Safe

- **Maximizing Profits:** To make money, telecom companies want to pack as many customers as possible onto the same physical servers.
- **The Security Threat:** Sharing server hardware means an infection in one cheap slice can slow down, spy on, or crash neighboring slices.
- **Why Buying More Servers Fails:** Giving every single customer their own separate physical hardware is 100% safe, but it is far too expensive and bankrupts the operator.
- **The Unsolved Question:** How can the network isolate infected customers in real time without wasting money on permanent, expensive hardware?

The Baseline Solution: 5Guard (IEEE 2026)

Foundation: 5Guard (Bolourian et al., IEEE TMC 2026)

- **Choose Your Security Level:** 5Guard introduced *Isolation Levels as a Service (ILaaS)*, allowing customers to choose how much protection they want when they sign up.
- **Tiered Pricing:** Customers pay different prices depending on whether they share servers with others or get dedicated hardware.
- **End-to-End Placement:** 5Guard uses mathematical optimization to place these slices across cell towers, transport networks, and cloud servers to maximize operator profit.

The Three Levels of Security in 5G

Level 0 (Shared)

Customers share the same processor and memory with everyone else. Lowest cost, but highest risk if someone gets hacked.

Level 1 (Virtual)

Puts a Virtual Machine around the customer to create a software wall. Moderate cost, and blocks basic malware from spreading.

Level 2 (Physical)

Gives the customer dedicated physical CPU cores that nobody else can touch. Maximum security, but highest hardware cost.

The Math Problem: The Giant Jigsaw Puzzle

A 5G network spans thousands of cell towers, optical links, and servers. 5Guard formulates this as a **Mixed-Integer Programming (MIP)** question with a vast number of independent variables:

Binary Variables

Discrete Yes / No Choices (0 or 1)

Which exact physical server, cell tower frequency PRB slot, and optical fiber wavelength hosts each slice.

Continuous Variables

Exact Sizing (≥ 0)

How much CPU compute power (MIPS) and network bandwidth capacity (Gbps) to allocate.

The Optimization Goal

Maximum Profit Under Security

Arrange all L0, L1, and L2 customers to satisfy requested security while maximizing total network profit.

Objective Function: **Maximize Profit = $\sum$ Revenue(r) – ($\gamma^{dep} + \gamma^{vio} + \gamma^{mig} + \gamma^{ovh}$)**

Profit = **Tenant Subscription Revenue** – (**Deployment Cost** γ^{dep} + **SLA Penalties** γ^{vio} + **Migration Cost** γ^{mig} + **Virtualization Overhead** γ^{ovh}).

5Guard: Finding the Best Setup

Solving this NP-hard Mixed-Integer Program (MIP) exactly takes too long for real-time 5G deadlines. So, 5Guard uses an ensemble of algorithms:

- **Exact Math (MIP Solver):** Uses SCIP to find the provably optimal allocation, but is slow on large networks.
- **Smart Heuristics (IAR / DTR):** Highly effective algorithms that balance isolation, revisitation, and speed.
- **Fast Heuristics:** Ultra-fast execution for strict, millisecond-level slice deadlines.

5Guard runs these methods in parallel and picks the best answer that finishes within the slice deadline.

The Big Problem with 5Guard

- **Blind to Hackers:** 5Guard only sets security when a customer joins, and never changes it.
- **No Defense:** If a customer asks for a cheap (L0) network and gets hacked later, 5Guard does nothing.
- **Wasting Money:** To be safe, companies just buy the expensive L2 security all the time, wasting huge amounts of money.

The Gap

The original authors said:

"Translating security requirements into isolation levels is left for future work."

5Guard Algorithm Ensemble: Trade-Off Analysis

Comparative evaluation of optimization methods running concurrently in the 5Guard parallel ensemble:

Algorithm	Strategy	Time	Quality	When It Wins
OPT (SCIP) Exact Optimal Solver	Branch-and-Cut MILP	High (100ms – mins)	100% (Optimal)	Small networks or generous deadlines
IAR Isolation-Aware Ranking	Greedy BFS + Isolation Ranking	Medium (20 – 50ms)	~96% (Near-Optimal)	Standard multi-tenant workloads
DTR Deterministic Rounding	Multi-path Latency Relaxation	Fast (10 – 30ms)	~90%	High network congestion & IoT slices
Fast / RNR Randomized Rounding	One-shot First-Fit	Ultra-fast (1 – 5ms)	~82% (Acceptable)	Emergency deadline timeout prevention

SCIP: Solving Constraint Integer Programs **MILP:** Mixed-Integer Linear Program **BFS:** Breadth-First Search

Operational Takeaway: Running these algorithms concurrently ensures the system never drops requests due to admission timeouts while capturing maximum possible profit.

Proposed Framework

Enter SecGuard

Risk-Adaptive Closed-Loop Isolation for Profit-Aware 5G Slicing

Research Scope: Three Core Objectives

1. Threat-to-Isolation Mapping

To detect 5G network attacks in real time and dynamically map compromised slices to L0/L1/L2 isolation escalation while preserving baseline customer SLAs.

[Completed Phase]

2. Predictive Admission Control

To formulate a reinforcement learning policy predicting downstream network impact to dynamically admit, reject, or defer slices for long-term profit optimization.

[Proposed Phase 2]

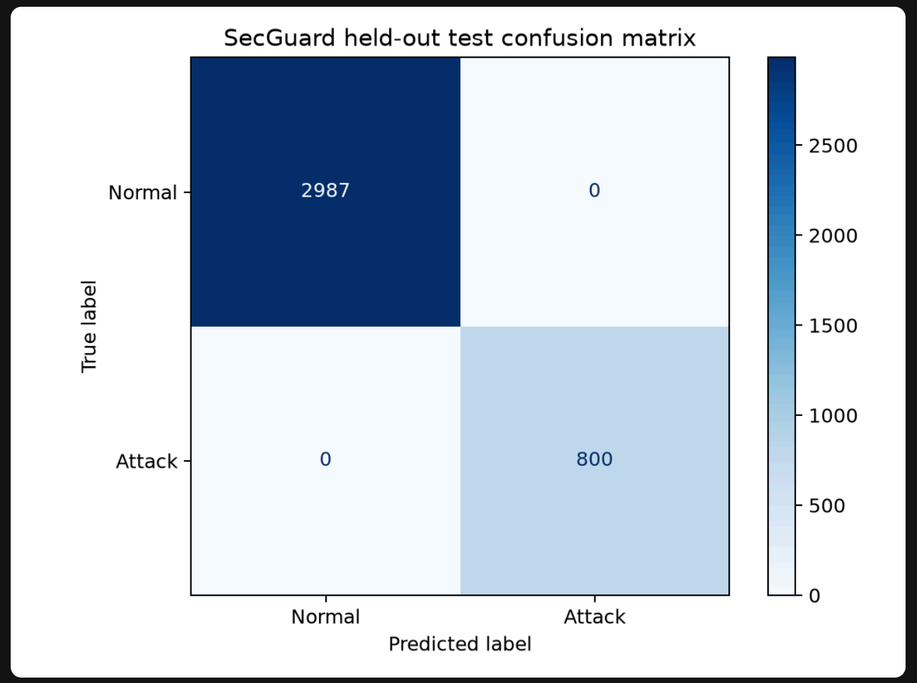
3. Meta-Algorithm Selection

To construct an intelligent meta-classifier routing requests between Exact (SCIP) and Heuristic solvers to satisfy strict sub-second slice deadlines.

[Proposed Phase 3]

Objective 1: System Architecture & Formulation

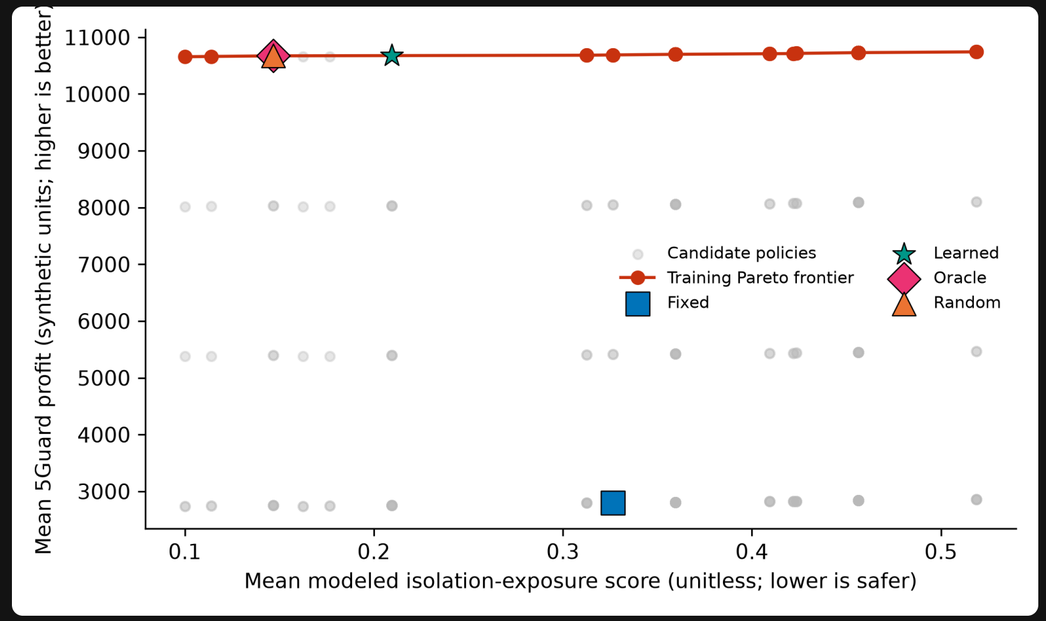
1. **Packet Telemetry Classification:** 300-tree Random Forest trained on 5G telemetry; achieves 100% test accuracy on held-out attack flows.
2. **Temporal Risk Aggregation:** Mathematical formulation aggregating sliding-window attack probabilities into a continuous slice risk metric (0.0 to 1.0).
3. **Risk-Adaptive Escalation:** Autonomous policy engine dynamically re-mapping slices to L1, L2, or Quarantine upon threshold breach.
4. **SLA Preservation Guarantee:** Provable non-downgrade constraint guaranteeing customer baseline SLAs are strictly maintained.



100% Test Accuracy on 5G Attack Patterns

Objective 1: ML-Based Policy Optimization

- **Optimization Challenge:** Determining optimal escalation thresholds balancing risk exposure against resource overhead.
- **Extensive Policy Exploration:** Simulated and evaluated 165 candidate threshold configurations across full-scale network topologies.
- **Surrogate ML Modeling:** Trained regression surrogate to construct the Pareto Frontier (Operator Profit vs. Exposure Proxy).
- **Optimal Operating Point:** Identified Pareto-optimal thresholds (0.30, 0.35, 0.90) maximizing profit under strict exposure constraints.



Surrogate Pareto Frontier for Threshold Selection

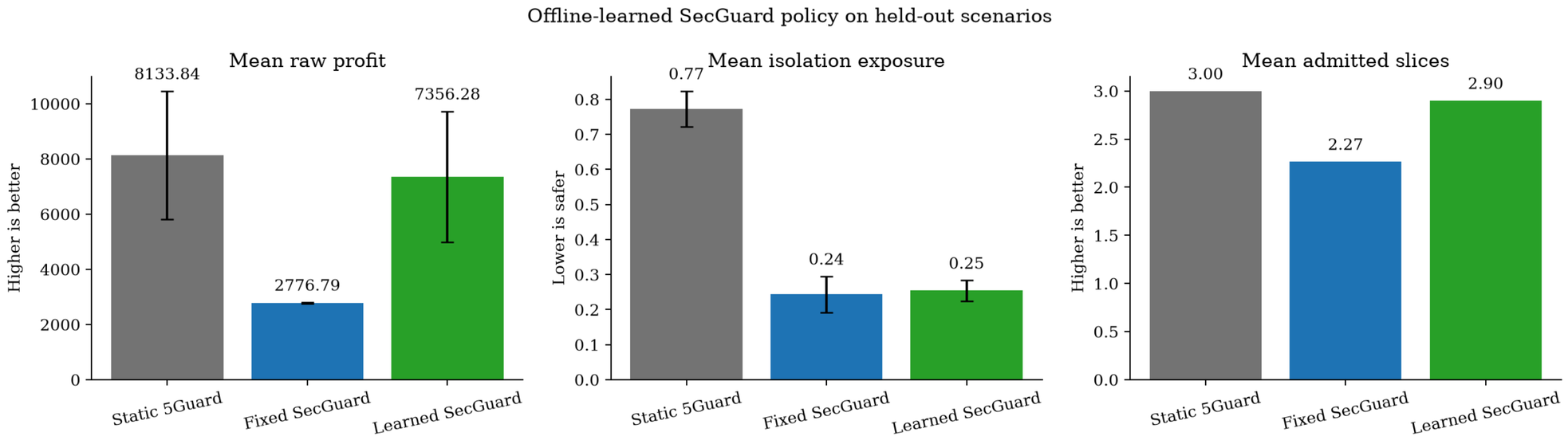
Objective 1: Performance Evaluation & Validation

Statistical validation evaluated across 30 diverse held-out network scenarios:

Metric	Baseline (No Security) Original 5Guard	Naive Security Fixed Rule Over-Quarantine	Proposed SecGuard ML-Optimized
Security Danger Proxy Lower is safer	0.77 (High Risk)	0.24	**0.25 (-67.1% exposure)**
Mean Network Profit Higher is better	$8,133	$2,776	**$7,356 (+164.9% recovery)**
Mean Slice Admission Out of 4 total requests	3.5 slices	2.2 slices	**2.9 slices**

Profit Calculation: Profit = ∑ Subscription Revenue − (Deployment Costs + SLA Penalties + Migration Costs + Virtualization Overheads)

Objective 1: Empirical Evaluation on Held-Out Scenarios



$7,356.28 (+164.9%)

Profit Recovery vs Fixed Security (Retains 90.4% unconstrained yield)

0.25 (-67.1%)

Isolation Exposure Proxy (Matches fixed security without over-quarantine)

2.90 Slices (+27.8%)

Tenant Slice Admission Rate (Eliminates false-positive slice rejection)

Conclusion & Research Roadmap

Key Research Findings

- **Profit Retention:** Retained **90.4%** ($7,356) of theoretical unconstrained profit while enforcing multi-tier slice isolation.
- **Risk Reduction:** Slashed network security exposure proxy by **67.1%** (0.25 vs 0.77) compared to static baseline slicing.
- **Profit Recovery:** Recovered **+164.9% profit** ($7,356 vs $2,776) compared to rigid rule-based over-quarantine.
- **Closed-Loop Feasibility:** Validated that ML-driven dynamic isolation is both computationally and commercially viable in 5G.

Proposed Future Roadmap

- **Phase 2: Learning-Based Admission Control**
  Formulate deep reinforcement learning (DRL) policies predicting long-term network impact to admit, reject, or defer slices under volatile demand.
- **Phase 3: Intelligent Algorithm Selection**
  Construct meta-learning classifier to dynamically route requests between Exact (MIP) and Heuristic solvers, satisfying strict sub-second deadlines.
- **Testbed Deployment:** Validate the framework on Open5GS/OAI software-defined radio testbeds.

References

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5. [5] Zuse Institute Berlin (ZIB), "SCIP: Solving Constraint Integer Programs," Mathematical Optimization Solver Suite, 2025. [Online]. Available: https://scipopt.org